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An outline of the natural force density method and its extension to quadrilateral elements

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A B S T R A C T

The Natural Force Density Method (NFDM) is a convenient method for finding configurations of membranes and funicular shell structures, providing viable equilibrium geometries in a single linear equilibrium analysis. It is an extension of the Force Density Method (FDM), first proposed by Linkwitz in 1971, as convenient alternative for finding configurations of cable nets, which became ubiquitous in the field of membrane design, serving as basis of several successful computer programs. However, the equivalence between a network of force density elements and a continuous membrane is far from obvious, and it may become quite dubious which force densities should be prescribed, to achieve a desired configuration, comprising a geometric shape and an associate stress field. This is seldom remarked in papers about the method, and even less aware of that might be the general users the of programs based on it. The NFDM, first proposed by Pauletti in 2006, treats directly the continuous problem, overcoming the difficulties of FDM to deal with irregular meshes. Nevertheless, the NFDM was originally proposed considering the triangular membrane element introduced by Argyris in 1974, and thus required a fully triangular tessellation of a reference domain. In this paper we present a more concise deduction and systematic outline of the NFDM, as well as a simple extension of the method to quadrangular elements, allowing the use of more general mesh generators for the initial references meshes, and selection of a broader class of membrane or shell finite elements, afterwards.

1. Introduction

Since Frei Otto’s pioneering works in the 1950′s, taut structures (encompassing cables, membrane and tensegrity structures) constitute an important research field in architecture and engineering. They are light, elegant and effective structures, whose applications range from large stadium roofs and high-rise building walls to pneumatic furniture or aerospace equipment. Cables and membranes do not withstand bending –and thus, neither compression–, therefore the shape of a taut structure cannot in general be imposed, but must be search, alongside with a corresponding stress field. The response of a taut structure to external loads then occurs around this configuration, where the structure is stiffened by the effect of the permanent prestressing loads. Since the initial configuration for load analysis is the final configuration obtained in the process of shape finding, we employ the terminology viable configuration to address simultaneously the initial equilibrium shape and its associated stress field (Pauletti and Pimenta, 2008). Once a viable configuration is found, it can be subsequently studied considering the generally nonlinear structural response of the structure to the design loads. In more recent years, the same shape-finding procedures have also become increasingly popular for finding bending-free, ‘funicular’ shapes for shell structures. Once a satisfactory funicular shape is found, stability and strength of the shell can be subsequently assessed by means of suitable structural analyses. In this paper, we encompass both taut structures and bending-free shells in the broader class of funicular structures, a well-established term in architecture and structural engineering (Shodek, 1980), (Adriaenssens et al., 2014).

A very convenient alternative method for finding viable configurations of funicular structures is the *force density method* (FDM) first proposed by Linkwitz (1971) and Schek (1974), in the context of cable nets. The method became also routinely applied to the shape finding of funicular surfaces, replaced by equivalent cable networks, which must be as regular as possible (otherwise it may become quite dubious which force densities should be prescribed to achieve a desired shape)

Pauletti (2006) proposed an extension of the FDM method for the shape finding of continuous membranes, based on the natural approach introduced by Argyris et al. (1974) for the Finite Element Method. The method, named *Natural Force Density Method*

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(NFDM), preserved the linearity of the original FDM. Afterward, Pauletti and Pimenta (2008) recognized that the imposition of natural force densities is equivalent to the imposition of 2ndPiola-Kirchhoff stresses on a reference configuration, a result previously pointed out by Bletzinger and Ramm (1999) and Wuechner and Bletzinger (2005) for the case of the original FDM. If the NFDM is applied iteratively it may converge to a minimal surface through a succession of viable configurations.

The NFDM was formulated on basis of Argyris' triangular membrane element and, accordingly, all previous publications on the method were limited to triangular tessellations. In this paper we provide further generality, presenting a simple extension of NFDM to quadrangular elements, based on the decomposition of the element quadrangular domain into four triangular sub-elements. We include some benchmarks to assess the performance of the new element for shape-finding, as well as its capability to ride out the bias that triangular meshes might introduce on the numerical results, depending on the mesh orientation. Moreover, we seek to present a systematic outline and a more concise deduction of NFDM, based exclusively in the equilibrium of forces, which highlights the fact that the NFDM is unmaterial, *i.e.*, the consideration of strains is completely unnecessary.

Since its proposition, the NFDM has found a relevant niche on the field of membrane designs, being independently incorporated into some computer programs, and referred by several independent researches, such as Pauletti (2010), Lee and Han (2011), Tsiatas and Katsikadelis (2011), Descamps et al. (2011), Gosling and Zhang (2012), Veenendaal and Block (2012), Tur et al. (2014), Yang et al. (2014), Greco et al. (2014), Kmet and Mojdis (2015), Kim et al. (2016), Li et al. (2016), Xu et al. (2015), Yee et al. (2015), Li et al. (2017), Mikula et al. (2015), Marmo and Rosati (2017), Tang and Li (2017), Veenendaal et al. (2017) Shimoda et al. (2018).

Gellin and Pauletti (2011, 2012, 2016) studied variable stress ratio conoids, finding excellent agreement of NFDM models with analytical solutions, detailed in Gellin and Pauletti (2013). Descamps et al. (2011) applies the FDM to the multicriteria optimization of lightweight bridge structures, explicitly referring to the weak form of the equilibrium equations proposed by Pauletti and Pimenta (2008). Koohestani (2014) presented a nonlinear force density method with a quadrilateral graph which closely resembles the NFDM quadrilateral element presented in the current paper, but without the element-wise approach adopted herein. Mikula et al. (2015) recognized the NFDM as one of a few algorithms able to produce structures that consistently approximate that minimal surfaces. Xu et al. (2015) proposed a modified nonlinear FDM for form-finding of aerospace antennas (MNFDM), comparing the original FDM with the NFDM and the Improved Nonlinear Force Density (INFDM) proposed by Xiang et al. (2010). Although the authors concluded that MNFDM presents a better performance, the proposed benchmarks in fact show fair agreement between these three last methods, with inferior results provided by the original FDM. Veenendaal and Block (2012) undertook a comprehensive comparison of form finding methods for general funicular networks, comprising stiffness matrix methods, force density methods (with several variations, including the NFDM), dynamic relaxation and particle-spring systems, remarking that Singer (1995) presented a triangular element which may be considered a precursor to the NFDM element. These authors recognized also the difficulties of prescribing force densities for general networks, pointing out that calibration of the force density values could be obtained by means of geometrical constraints, in an optimization problem. They also recognized that a cable-net analogy could not be straightforwardly applied to shape finding of minimal surfaces. Several meshes were tested on simple saddle and conoidal networks, although only with rather regular meshes, so the difficulties related to irregular meshes, which we address in the current paper, were largely circumvented there.

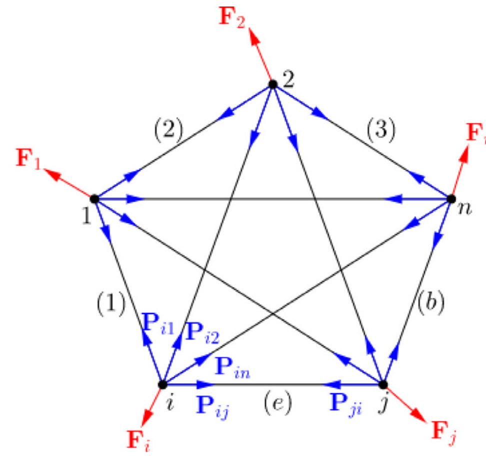

Fig. 1. A system of central forces.

Calibration of force densities to the shape finding of simple geometries have been undertaken by several authors (Zhang and Ohsaki, 2006), (Koohestani, 2012) and (Koohestani and Guest, 2013), (Cercadillo-Garcia and Fernández-Cabo, 2016), (Quagliaroli and Malerba, 2013), (Greco and Cuomo, 2012). However, when complex and irregular geometries come into play, limitations of the FDM becomes evident. Veenendaal et al. (2017) applied the FDM to the shape finding of a flexibly formed, mesh-reinforced concrete shell roof. The geometry of the network reinforcement was obtained by an evolutionary optimization procedure, by which the force densities on single members were defined by interpolation of a set of prescribed force density values, limiting the ration between allowable force densities to 1:20, "to create reasonable shapes without too abrupt changes in curvature and resulting forces". The procedure was prone to curve the mesh on itself at the region of the supports. That was remedied by "calculating force densities of the network's triangulated projection, using the linear NFDM". This is evidence on the advantage of the NFDM to cope with irregular meshes, somehow contradicting previous statement by these authors that FDM and its variations are equivalent (Veenendaal and Block, 2012). In a basic sense, of course they are, since all of them simply express force equilibrium under funicular states, but in practice the NFDM is superior to cope with continuous surfaces and irregular meshes, as we endeavor to show in the sequel.

2. Equilibrium of cable networks and the force density method

We initially review the original FDM, considering the problem of the equilibrium of a cable network, or more generally, a system of central forces, as depicted in Fig. 1. We store the Cartesian coordinates of nodes $i = 1, 2, \ldots, n$ in column-matrices $\mathbf{x}_i$, which are collected in a *global position vector* $\mathbf{X} = [\mathbf{x}_1^T \;\; \mathbf{x}_2^T \;\; \ldots \;\; \mathbf{x}_n^T]^T$ (the double transposition might seem awkward, but allows a more compact inline notation). Similarly, external and internal forces at the nodes are stored in column-matrices $\mathbf{F}_i$ and $\mathbf{P}_i = -\sum_{j=1}^{n} N_{ij}\mathbf{v}_{ij}$ (where $\mathbf{v}_{ij} = (\mathbf{x}_j - \mathbf{x}_i)/\|\mathbf{x}_j - \mathbf{x}_i\|$) and are grouped in *external and internal global load vectors* $\mathbf{F}$ and $\mathbf{P}$, with analogous definitions.

The problem of equilibrium can then be posed as that of finding a configuration $\mathbf{X}^*$ such that

$$\mathbf{P}(\mathbf{X}^*) - \mathbf{F} = \mathbf{0}, \tag{1}$$

where the external forces $\mathbf{F}$ are hereafter considered as independent of $\mathbf{X}$, without loss of generality.

We consider that a generic element identified by the index e connects nodes i and j and is under a normal load N_e. The element vector position can be extracted from the global position vector according to $\mathbf{x} = [\mathbf{x}_1^T \quad \mathbf{x}_2^T]^T = \mathbf{AX}$, where $\mathbf{A}$ is the order $6 \times 3n$ Boolean *incidence matrix* of that element, such $\mathbf{A}_{1i} = \mathbf{A}_{2j} = \mathbf{I}_3$and $\mathbf{A}_{1k} = \mathbf{A}_{2k} = \mathbf{0}_3$, $k \neq i$, $k \neq j$, where $\mathbf{0}_3$ and $\mathbf{I}_3$ are, respectively, the null and identity matrices of order three. It can be readily verified that the global internal load vector is assembled according to

$$\mathbf{P} = \sum_{e=1}^{n_e} \mathbf{A}_e^T \mathbf{p}_e, \tag{2}$$

where index e spans all n_e elements of the system, and where the element *internal force vector* is successively rewritten as

$$\mathbf{p} = N\begin{bmatrix} -\mathbf{v} \\ \mathbf{v} \end{bmatrix} = \frac{N}{\|\mathbf{x}_j - \mathbf{x}_i\|}\begin{bmatrix} -\mathbf{x}_j + \mathbf{x}_i \\ \mathbf{x}_j - \mathbf{x}_i \end{bmatrix} = \frac{N}{\ell}\begin{bmatrix} -\mathbf{x}_j + \mathbf{x}_i \\ \mathbf{x}_j - \mathbf{x}_i \end{bmatrix}$$
$$= n\begin{bmatrix} -\mathbf{x}_j + \mathbf{x}_i \\ \mathbf{x}_j - \mathbf{x}_i \end{bmatrix}, \tag{3}$$

where index e was dropped out, for the sake of concision. Moreover, we have implicitly defined the *element unit vector* $\mathbf{v}$, *element length* ℓ and *element force density* $n = N/\|\mathbf{x}_j - \mathbf{x}_i\|$.The internal load vector $\mathbf{P}$ may vary as function of nodal coordinates $\mathbf{X}$ and the equilibrium problem (1) may be quite nonlinear. However, imposing constant values for the element force densities, we arrive at a system of linear equilibrium equations

$$\left(\sum_{e=1}^{n_e} \mathbf{A}_e^T \mathbf{k}_e \mathbf{A}_e\right)\mathbf{X} = \mathbf{KX} = \mathbf{F}, \tag{4}$$

where we recognize the element force *density stiffness matrix*,

$$\mathbf{k} = n\begin{bmatrix} \mathbf{I}_3 & -\mathbf{I}_3 \\ -\mathbf{I}_3 & \mathbf{I}_3 \end{bmatrix}. \tag{5}$$

Of course, an optimized implementation should consider the symmetry and sparsity of this matrix. During shape finding, it is often assumed $\mathbf{F} = \mathbf{0}$, and some components of $\mathbf{X}$ must be imposed, to avoid the trivial solution, $\mathbf{X} = \mathbf{0}$.

Fig. 2 illustrates some cable nets whose shapes were found using the procedure outlined above. Both initial and final geometries are show, although essentially only the coordinate of the four vertices are be required, if a set of force densities is given. However, in this example we used the initial geometry to calculate the force densities according to the initial length of the members and prescribed initial normal loads. Three different geometries were generated, prescribing a uniform load the $N_i = 1$ acting on the internal members and three different values for the normal loads acting on the border elements ($N_b = 1$, $N_b = 5$, $N_b = 30$). Once solved for $\mathbf{X}$, the final normal loads acting on the members are calculated considering their final length and the prescribed force densities. It is seen that non-uniform normal forces are obtained at the final, viable configurations, and that elements with higher normal loads become longer. These are intrinsic characteristic of the FDM, that the designer has no full control on the resulting normal forces or the length of the cables. Nonetheless, some level of control is possible, if one reimposes forces on the resulting geometry, recalculates the corresponding force densities, and iterates the method.

For regular meshes as those of Fig. 2, a network of FDM linear elements can offer satisfactory results for an equivalent membrane, with little computation cost. However, when irregular meshes are considered, such as the one shown in Fig. 3(a), it is no trivial matter to prescribe a good set of force densities on a network of linear elements, to represent an equivalent membrane surface. Fig. 3(b) and (c) show the geometry obtained by imposing a uniform normal load $N_i = 1$ to the inner members and $N_b = 30$ to the border members, with the same boundary conditions as in the case of the networks of Fig. 3. Once again, force densities are calculated considering the initial length of the members. Distortion of the mesh at the resulting configuration is quite apparent, and the quality of the overall geometry is also degraded, especially along the borders. Moreover, correlation between forces on the network members and stresses on an equivalent surface is far from evident.

3. The natural force density method

We now consider a triangular membrane element, depicted in Fig. 4(a), in an initial configuration Ω_0, at which it is already under an initial stress field $\boldsymbol{\sigma}_0$. We seek knowledge of the element's final configuration Ω and the final stress field $\boldsymbol{\sigma}$.

Element nodes and edges are numbered such that edges are facing nodes of same number, as shown in Fig. 4(b). Nodal coordinates are referred to a global Cartesian system, and a local coordinate system, indicated by an upper hat, is adapted to every element configuration, such that the $\hat{x}$ axis is always aligned with edge 3, oriented from node 1 to node 2, whilst the $\hat{z}$ axis is normal to the element plane. The element position vector is given by $\mathbf{x} = [\mathbf{x}_1^T \quad \mathbf{x}_2^T \quad \mathbf{x}_3^T]^T$, where $\mathbf{x}_i$, $i = 1, 2, 3$ are the nodal coordinates. The lengths of the element edges can then be computed by $\ell_i = \|\mathbf{x}_k - \mathbf{x}_j\|$, and unit vectors parallel to element edges are denoted by $\mathbf{v}_i = (\mathbf{x}_k - \mathbf{x}_j)/\ell_i$, while unit vectors normal to the element sides are given by $\mathbf{n}_i = -\mathbf{k} \times \mathbf{v}_i$, with indexes $i, j, k = 1, 2, 3$ in cyclic permutation, and $\mathbf{k}$ is a unit vector, normal to the element surface Ω. We consider a prescribed uniform plane stress tensor $\boldsymbol{\sigma}$ and collect its independent components in the local coordinate system in a pseudo-vector, $\hat{\boldsymbol{\sigma}} \equiv [\sigma_{\hat{x}} \quad \sigma_{\hat{y}} \quad \tau_{\hat{x}\hat{y}}]^T$. Side stress vectors are given by $\boldsymbol{\rho}_i = \boldsymbol{\sigma}\mathbf{n}_i$ and integration along the element sides provides the element internal load vector $\mathbf{p}$, according to

$$\mathbf{p} = \begin{bmatrix} \mathbf{p}_1 \\ \mathbf{p}_2 \\ \mathbf{p}_3 \end{bmatrix} = \frac{t}{2}\begin{bmatrix} \ell_2\boldsymbol{\rho}_2 + \ell_3\boldsymbol{\rho}_3 \\ \ell_1\boldsymbol{\rho}_1 + \ell_3\boldsymbol{\rho}_3 \\ \ell_1\boldsymbol{\rho}_1 + \ell_2\boldsymbol{\rho}_2 \end{bmatrix}. \tag{6}$$

We can insert the contributions of each element to the global equilibrium Eq. (1), considering Eq. (2) and suitable $9 \times 3n$ Boolean incidence matrices.

Referring to Fig. 5, the vector of internal forces can also be decomposed into components parallel to the elements' sides, or "*natural loads*" N_i, such that

$$\mathbf{p} = \begin{bmatrix} \mathbf{p}_1 \\ \mathbf{p}_2 \\ \mathbf{p}_3 \end{bmatrix} = \begin{bmatrix} N_2\mathbf{v}_2 - N_3\mathbf{v}_3 \\ N_3\mathbf{v}_3 - N_1\mathbf{v}_1 \\ N_1\mathbf{v}_1 - N_2\mathbf{v}_2 \end{bmatrix}. \tag{7}$$

We collect the natural loads in a vector $\mathbf{N} = [N_1 \quad N_2 \quad N_3]^T$ and compare (7) to (6), to obtain, after some algebra (see Appendix),

$$\mathbf{N} = V\boldsymbol{\mathcal{L}}^{-1}\mathbf{T}^{-T}\hat{\boldsymbol{\sigma}}, \tag{8}$$

where $V = A \times t$ is the element volume, $\boldsymbol{\mathcal{L}} = diag\{\ell_1, \ell_2, \ell_3\}$ is a diagonal "*length matrix*" and $\mathbf{T}^{-T}$ is a transformation matrix, whose transpose-inverse is given by

$$\mathbf{T} = \begin{bmatrix} \cos^2\gamma & \sin^2\gamma & -\sin\gamma\cos\gamma \\ \cos^2\beta & \sin^2\beta & \sin\beta\cos\beta \\ 1 & 0 & 0 \end{bmatrix}. \tag{9}$$

We choose to display $\mathbf{T}$ rather than $\mathbf{T}^{-T}$, because the first has a more compact analytical expression. We also remark that this somewhat awkward notation keeps compatibility with previous works, in which Eq. (8) was deduced considering the principle of virtual work, and involving the definition of some natural strains and stresses. However, the NFDM is unmaterial, and definition of natural strains or stresses is altogether unnecessary for deduction of Eq. (8), as shown in the Appendix.

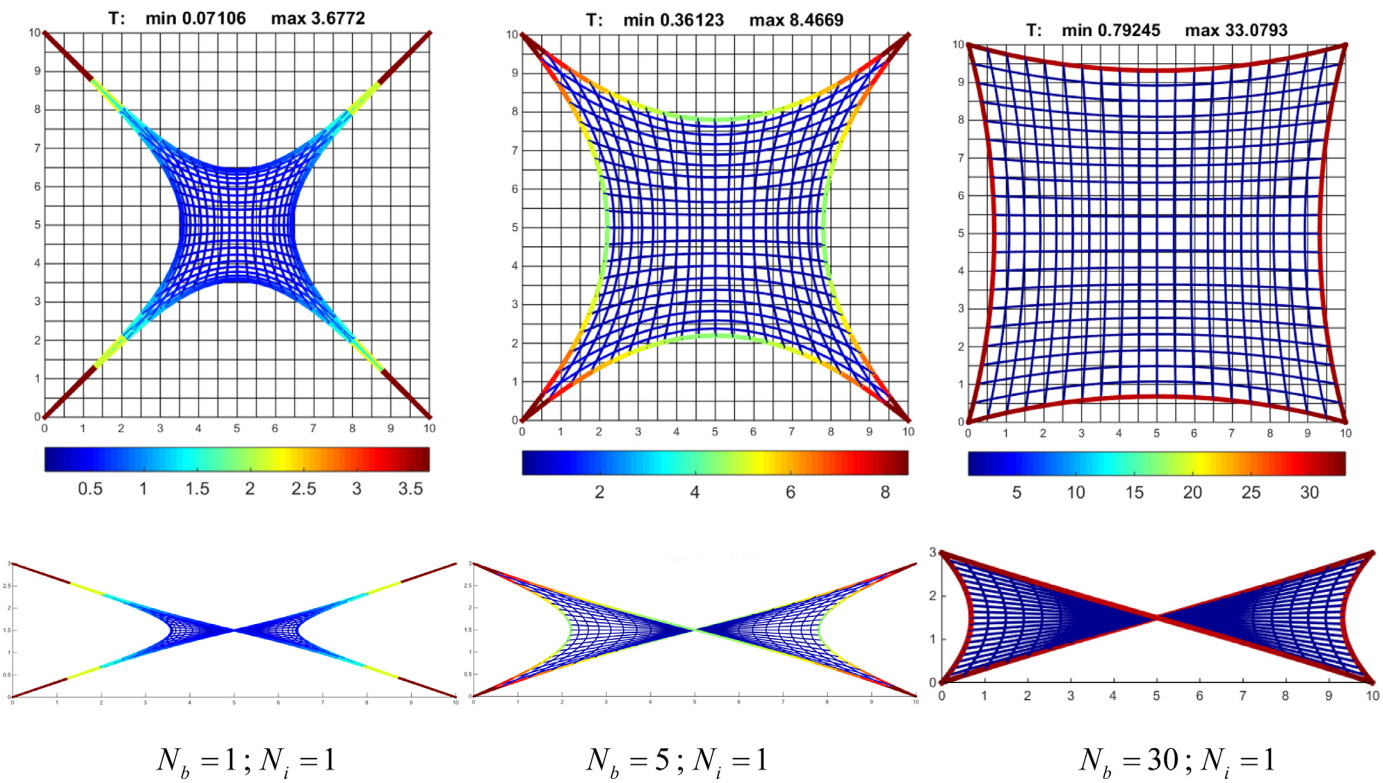

Fig. 2. Some cable net shapes found by the FDM, with different ratios between the initial normal load at the border cables (N_b), and those at the inner cables (N_i).

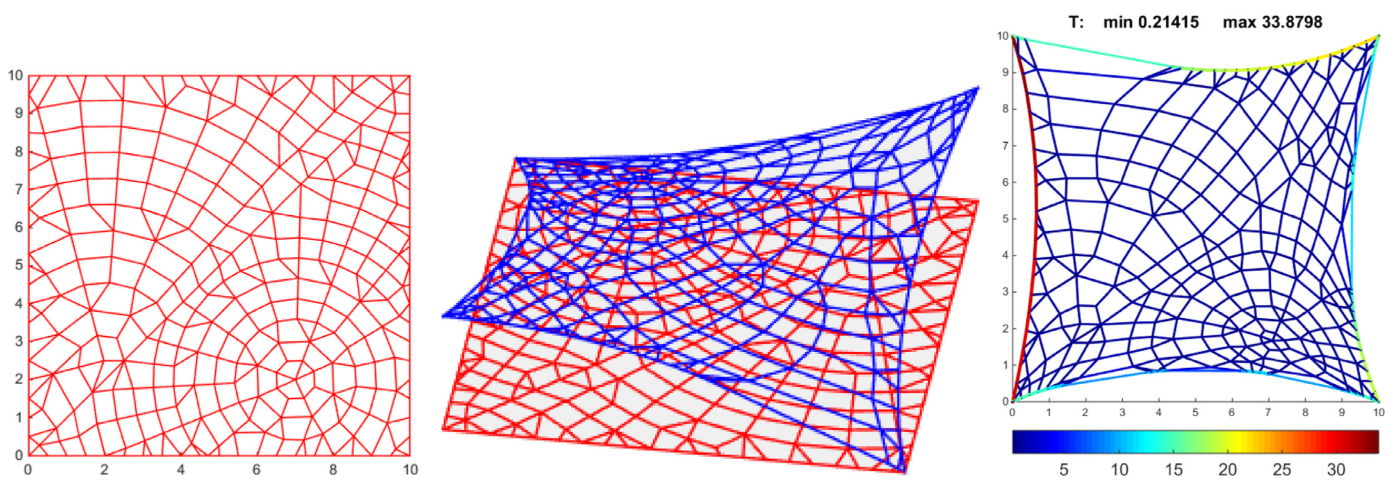

Fig. 3. (a) an irregular network, for which it is not trivial to define good force density values; (b/c) network geometry obtained by imposing $N_b = 30$; $N_i = 1$ and vertical displacements to two opposed vertices.

Comparison of Eq. (7) with Eq. (3) also suggests the definition of some *natural force densities*$n_i = N_i / \ell_i$ in such a way that Eq. (7) can be rewritten as

$$\mathbf{p} = \mathbf{k}\mathbf{x} \tag{10}$$

where, similarly to the line FDM element, we recognize the *force density stiffness matrix*

$$\mathbf{k} = \begin{bmatrix} (n_2 + n_3)\mathbf{I}_3 & -n_3\mathbf{I}_3 & -n_2\mathbf{I}_3 \\ -n_3\mathbf{I}_3 & (n_1 + n_3)\mathbf{I}_3 & -n_1\mathbf{I}_3 \\ -n_2\mathbf{I}_3 & -n_1\mathbf{I}_3 & (n_1 + n_2)\mathbf{I}_3 \end{bmatrix}. \tag{11}$$

It turns out that the stiffness matrix of a membrane force density element is equivalent to the assembling of three linear FDM elements, whose densities are collected in a *vector of natural force densities*

$$\mathbf{n} = \begin{bmatrix} n_1 \\ n_2 \\ n_3 \end{bmatrix} = \begin{bmatrix} \frac{N_1}{\ell_1} & \frac{N_2}{\ell_2} & \frac{N_3}{\ell_3} \end{bmatrix}^T = \boldsymbol{\mathcal{L}}^{-1}\mathbf{N} = V\boldsymbol{\mathcal{L}}^{-2}\mathbf{T}^{-T}\hat{\boldsymbol{\sigma}}. \tag{12}$$

When force densities are directly prescribed, the NFDM depends exclusively on topology, rather than on geometry. However, direct prescription of the natural force densities $\mathbf{n}$ for each element may be complicated, especially in the case of irregular meshes. It is quite more intuitive to prescribe an initial stress field $\hat{\boldsymbol{\sigma}}_0$, acting on a reference configuration Ω_0, and compute the associated force densities according to

$$\mathbf{n}_0 = V_0\boldsymbol{\mathcal{L}}_0^{-2}\mathbf{T}_0^{-T}\hat{\boldsymbol{\sigma}}_0. \tag{13}$$

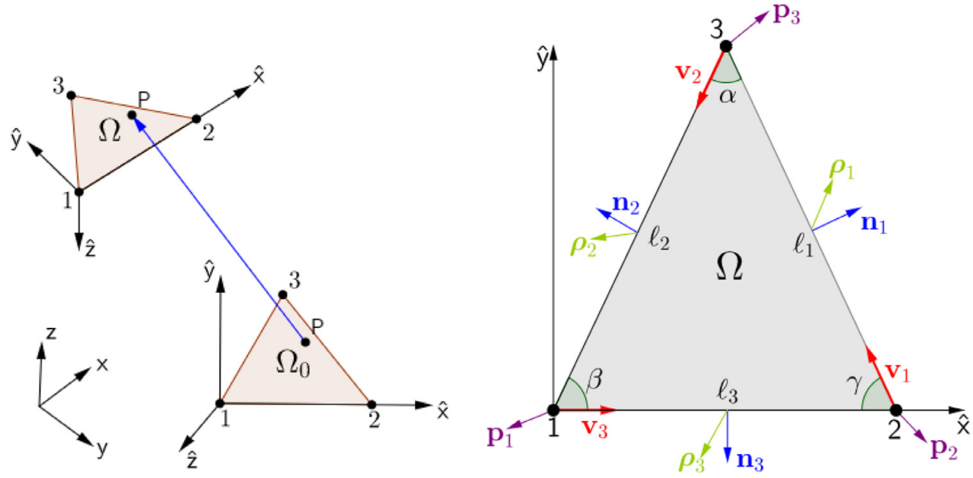

Fig. 4. (a) A triangular element in two different configurations; (b) Definitions of element quantities.

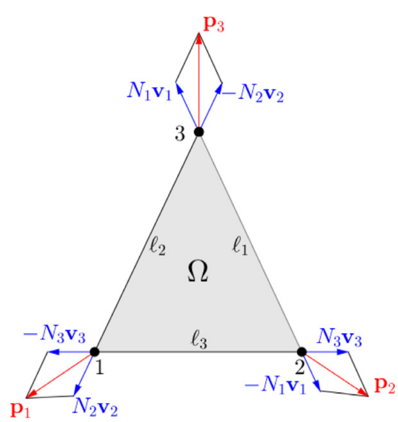

Fig. 5. Internal nodal forces decomposed into components parallel to the element sides.

Once the solution $\mathbf{X}$ is obtained, the associated viable stress field is given by

$$\hat{\boldsymbol{\sigma}} = V^{-1}\mathbf{T}\mathcal{L}^2\mathbf{n}_0 \tag{14}$$

where V, $\mathbf{T}$, $\mathcal{L}$ refer to the element geometry at the final equilibrium configuration.

It can be shown that $\boldsymbol{\sigma}_0$ acting on Ω_0 corresponds to the 2nd Piola-Kirchhoff stresses (2PK) associated to the final Cauchy stresses $\boldsymbol{\sigma}$acting on the equilibrium configuration (Bletzinger and Ramm, 1999), (Wuechner and Bletzinger, 2005), (Pauletti and Pimenta, 2008). References Pauletti (2006), Pauletti and Pimenta (2008) and Pauletti (2011) discuss the application of the NFDM to the shape finding of general membrane surfaces. The method has also been successfully applied to shape finding of bending-free shell surfaces (Souza and Pauletti, 2016).

The NFDM provides viable configurations (defined by the pair ($\boldsymbol{\sigma}$, $\mathbf{X}$)) in a single step, if no restrictions are imposed to the associated stress field. This is indeed enough for many practical applications. Moreover, if applied iteratively, the method provides convergence to minimal surfaces, re-imposing uniform and isotropic 2PK stresses to the shapes obtained at every iteration, in such a way that a uniform and isotropic Cauchy stress field is reached through a succession of viable configurations. This is a clear advantage, if compared to nonlinear methods, which may also converge to a minimal solution, but usually through a series of non-equilibrium configurations.

The NFDM can also be applied to the shape finding of non-minimal surfaces, by imposing non-isotropic 2PK stress fields. In that case, however, even though a viable shape can still be obtained at every linear step, there is no guarantee that an arbitrarily prescribed, non-isotropic Cauchy stress field can be achieved. Furthermore, since geometry varies during iterations, definition of stress directions becomes more complicate. A convenient way to prescribe a non-isotropic stress field on an NFDM element is to consider of a director plane Π, whose intersection with the element's surface Ω provides the direction of one of the principal 2PK stresses acting on the element. Fig. 6 shows how a horizontal plane may serve as director for a straight conoid, while a vertical plane is adequate to a saddle surface. Being $\mathbf{n} \perp \Pi$ and ($\mathbf{i}, \mathbf{j}, \mathbf{k}$) the unit vectors of the element local coordinate system, with $\mathbf{k} \perp \Omega$, the principal stress directions are given by unit vectors $\mathbf{i}' = (\mathbf{k} \times \mathbf{n})/\|\mathbf{k} \times \mathbf{n}\|$ and $\mathbf{j}' = \mathbf{k} \times \mathbf{i}'$, which are rotated with respect to the element local coordinate system by an angle $\theta = \arcsin((\mathbf{i}' \times \mathbf{i}) \cdot \mathbf{k})$.

The NFDM has been implemented in *ix-Cube 4.10*, a foremost commercial software for membrane design, becoming one of its

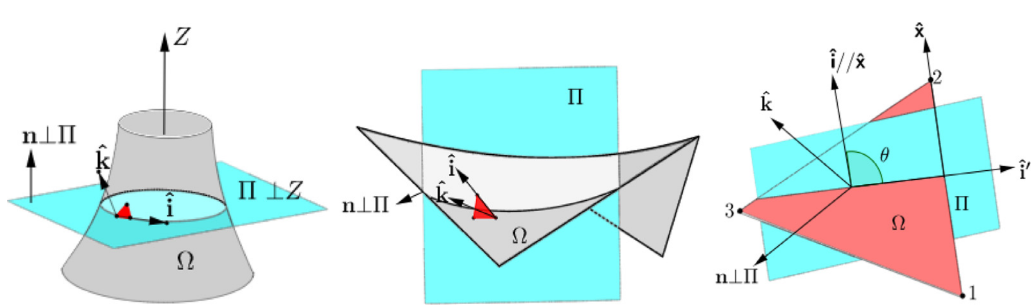

Fig. 6. Definition of principal stress directions using convenient director planes.

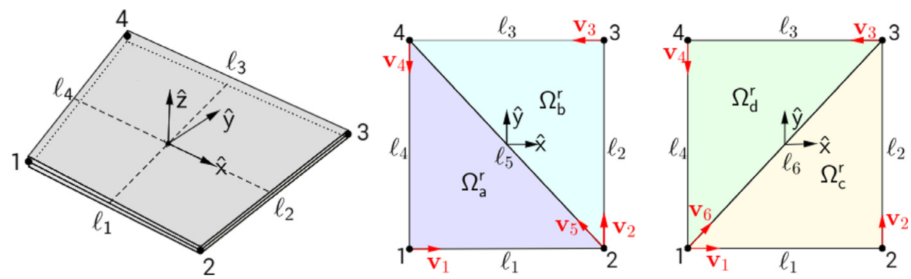

Fig. 7. (a) a quadrangular element; (b/c) two alternative partitions of the quadrangular element into two triangular elements.

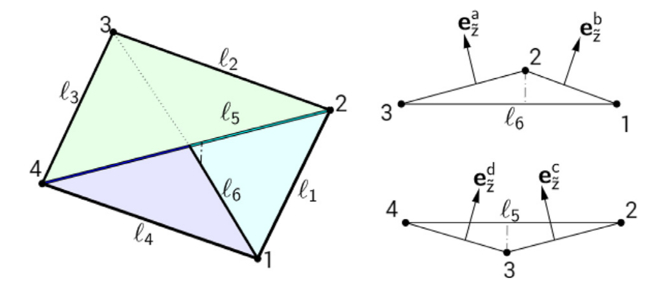

Fig. 8. Non-conforming subdivisions of the quadrangular element into two layers of triangular elements.

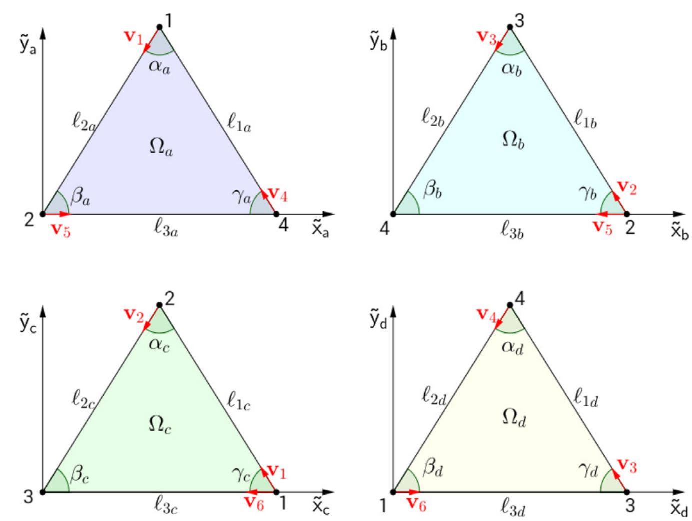

Fig. 9. Local coordinates systems on each part of the element (indicated by upper tildes).

standard methods for shape finding. Tan and Pauletti (2016) compared the performance of different shape finding methods available in this program, finding out that NFDM has a better performance with respect to the alternative methods available in the same program, presumably implemented with even programming skills.

4. Extension of the NFDM to quadrangular elements

An important advantage of the NFDM compared to the original FDM, for the shape finding of membrane structures, is that NFDM can be used with non-uniform meshes, whilst an FDM mesh ought to be as regular as possible, otherwise it may become dubious which force densities should be prescribed to the equivalent network, to achieve a desired shape. Furthermore, once a shape is found by FDM, the resulting forces in the network must be converted into equivalent surface stresses, as a function of some 'influence width' for each line element, resulting in crude estimative of the stresses on the equivalent membrane. Besides, even if both FDM and NFDM requires some iterative procedure to cope with external forces such as transversal pressure or self-weigh (whose definition depends of the geometry, a priori unknow), the NFDM does not require any *ad hoc*, mesh dependent procedure to compute these forces at every iteration.

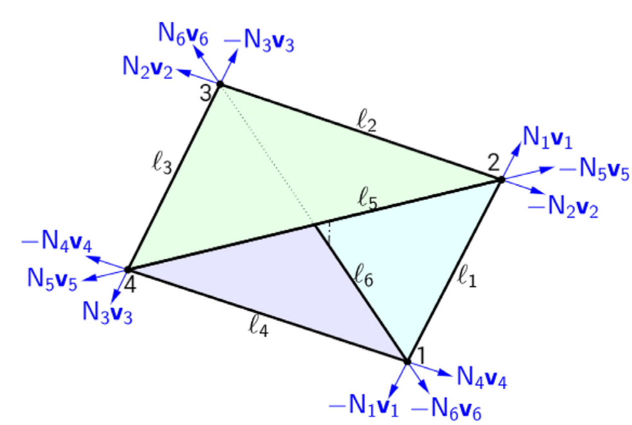

Fig. 10. Natural forces in the quadrangular element.

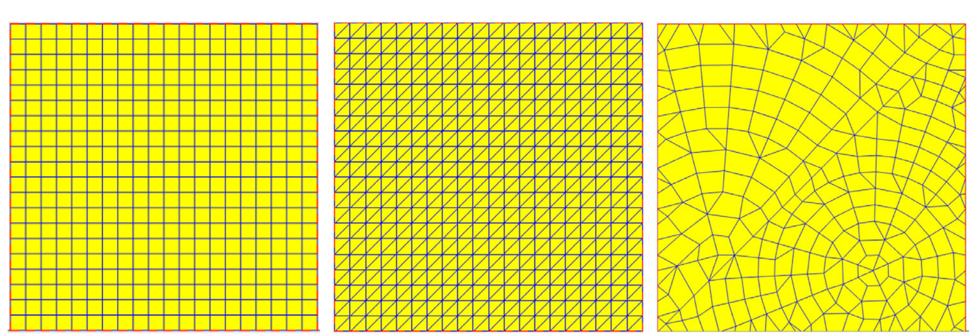

Fig. 11. A flat square domain tessellated with: *(a)* a set of quadrangular elements ('Model 1); *(b)*, a set of triangular elements with the same set of nodes as in Model 1, corresponding to 'Model 2' (as it is) and to 'Model 3' (reflected); *(c)* a set of irregular triangular and quadrangular elements ('Model 4').

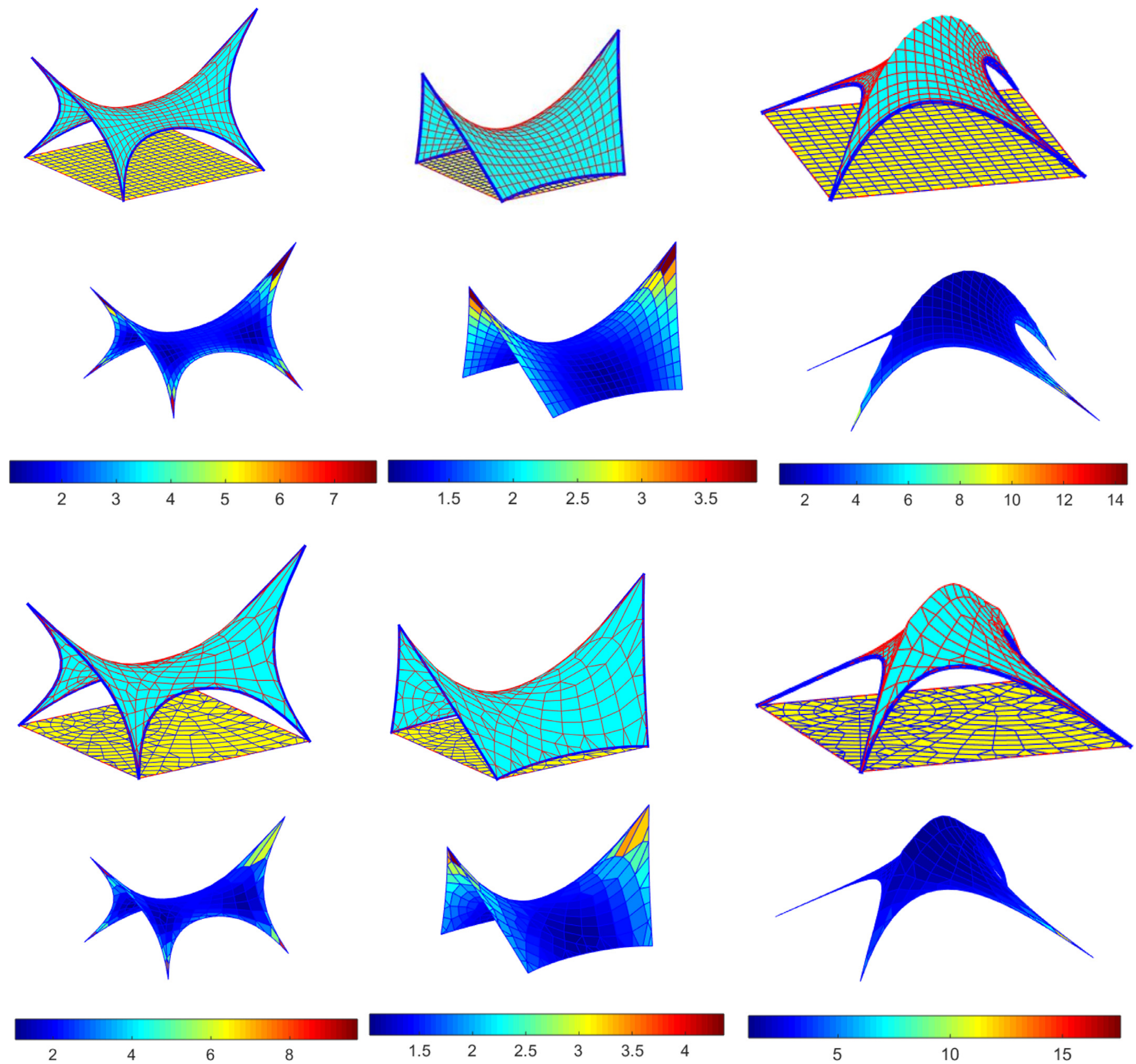

Fig. 12. Viable configurations generated in one iteration of the NFDM, through the imposition of different sets of nodal displacements and border force densities to a rectangular reference domain. Colors represent the magnitude of the first principal stresses (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.).

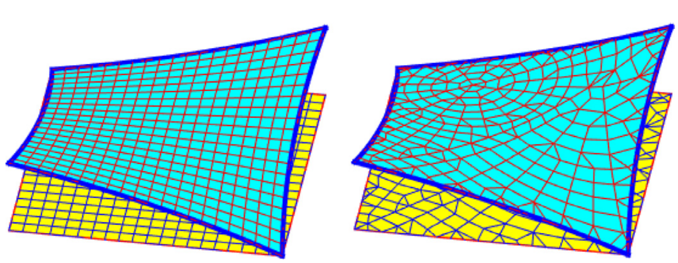

Fig. 13. Viable configurations for an initial isotropic stress field, found with Models 1 and 4 and a single NFDM step.

In this paper, we provide further generality to the NFDM, with respect to previous works, by extending it to quadrangular elements, which are indeed preferred over triangular elements, by many mesh generator and structural analysis systems, since quadrangular elements usually provided better performance, and triangular structural elements tessellation may introduce biased stress results. Quadrangular elements will also be more suited to model anticlastic surfaces with rougher meshes. In fact, as remarked by Remacle et al. (2012) the discussions about if and why quadrilaterals are better than triangles remained passionate in the finite element community. Nonetheless, in recent years, with the advent of spline-based surface representations, quadrilaterals regained favor over triangular representations, due to their tensor-product nature, capability to be aligned to principal curvature directions, and easy integration with CAD. Availability of quadrangular elements is also important for mesh legacy, and even superior

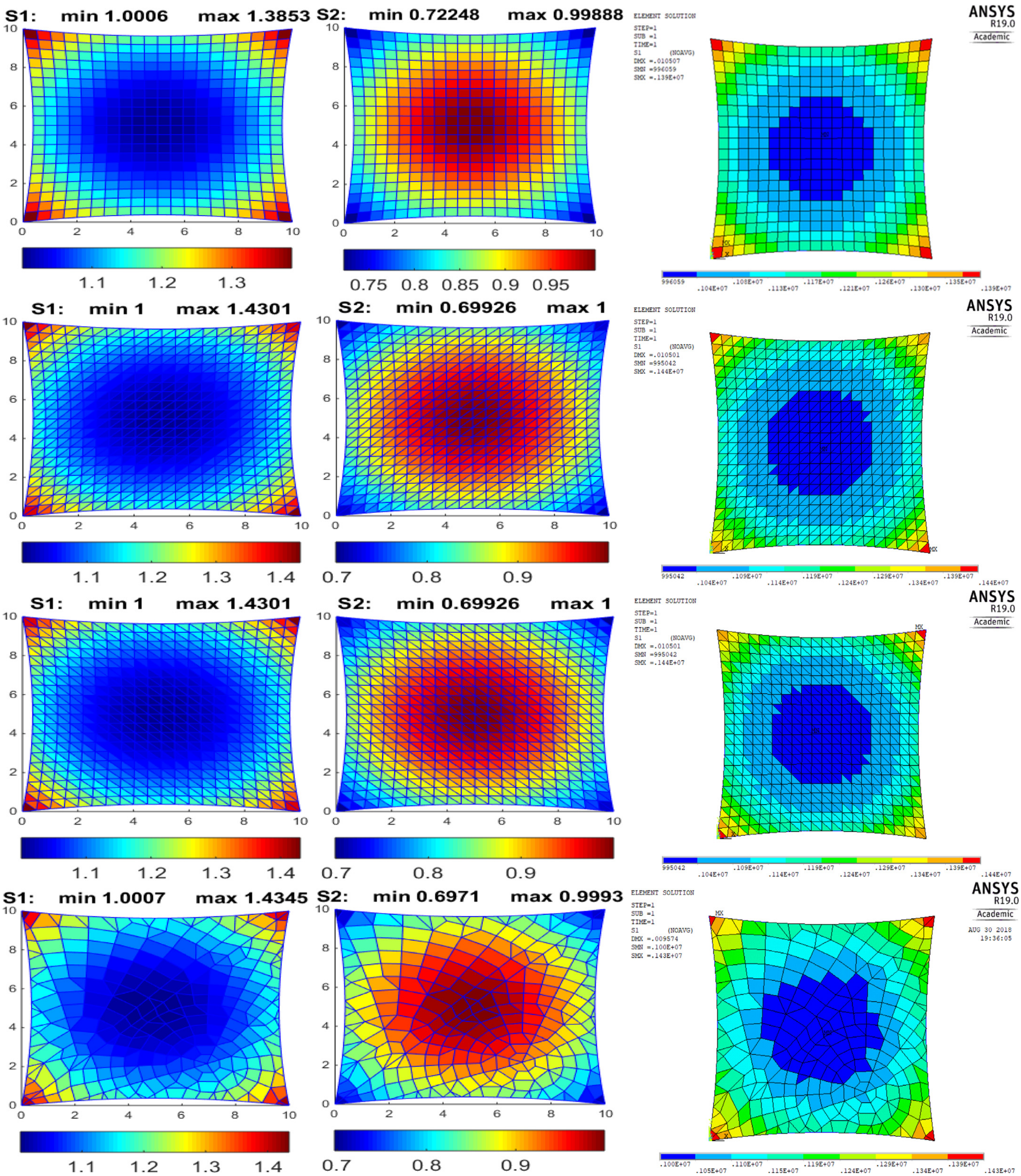

Fig. 14. Stress fields results for Models 1 to 4 (from line 1 to 4) after a single NFDM step . 1st column: 1st principal stresses σ_I; 2nd column: 2nd principal stress σ_{II}; 3rd column: 1st principal stress $\sigma_I$obtained with Ansys.

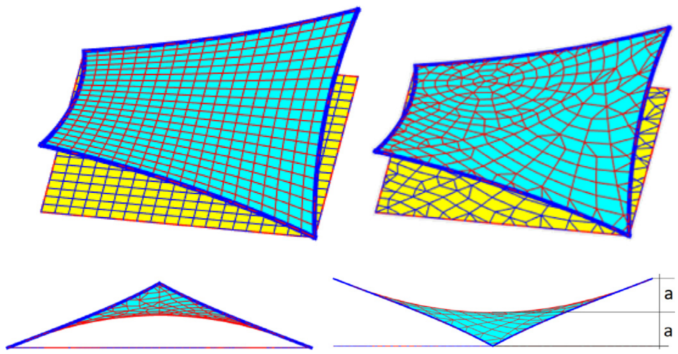

Fig. 15. A minimal surface obtained after 10 steps of the NFDM. The saddle point is at mid-height, *i.e.* $a = \frac{1}{2}u_z = 1.5m$.

mesh aesthetics may be claimed. Quadrilateral elements and mesh generators remain active fields in engineering and CAD researches (Daniels et al., 2011), (Remacle et al., 2012), (Bommes et al., 2013), (Rathod et al., 2014), (Araújo and Celes, 2014) (Surynková, 2016), (Zhong and He, 1998), (Zhou et al., 2018). As a matter of fact, several important engineering applications make use of quadrilateral meshes, thus availability of quadrilateral elements might prove resourceful in many practical instances.

Although some preliminary results on the extension of NFDM to quadrangular element were anticipated in Fernandes and Pauletti (2016), in this paper we recast it within the more systematic framework outlined in the previous sections. We consider the quadrangular element shown in Fig. 7(a), which may be flat or a hyperbolic paraboloid. Axis $\hat{x}$ connects the centroid to the midpoint of side 2, with a base vector$\mathbf{e}_{\hat{x}}$. Base vector $\mathbf{e}_{\hat{z}}$is obtained by the cross-product of $\mathbf{e}_{\hat{x}}$ and a vector connecting the centroid to the middle of side 3. Finally, $\mathbf{e}_{\hat{y}} = \mathbf{e}_{\hat{z}} \times \mathbf{e}_{\hat{x}}$. With these definitions, we keep compatibility with Ansys' SHELL181 element, which will be used further up for benchmarking.

Fig. 7(b) and (c) show two alternative partitions of the quadrangular element into triangular parts, without introduction of extra nodes, but resulting in different element surfaces, thus artificially biasing the element structural behavior. Therefore, we superimpose both alternative partitions, with half the element thickness for each one, as shown in Fig. 8. On doing so, we overcome the structural biasing, but a gap is introduced between the two layers, which might complicate the estimative of element stresses. Nonetheless, because the smooth surface of a nonplanar quadrangular element is bounded by the non-smooth surfaces of the alternative triangular layers, and both these layers provide convergent results, in terms of displacements and stresses, it is intuitive that their superposition, with half the thickness assigned for each one, will also converge, as in fact is confirmed by the examples we present in the sequel. This strategy somewhat resembles the Smoothed Finite Element Method proposed by Dai, Liu and Nguyen for plate and shell problems (Dai et al., 2007), (Liu et al., 2007), although those authors avoid nonconformity, by introducing an internal node, whose stiffness is lumped to the external ones. In this paper, however, we limit ourselves to a heuristic validation of our strategy, through its application to some selected problems, as discussed in next session.

$$\mathbf{k} = \begin{bmatrix} (n_1+n_4+n_6)\mathbf{I}_3 & -n_1\mathbf{I}_3 & -n_6\mathbf{I}_3 & -n_4\mathbf{I}_3 \\ -n_1\mathbf{I}_3 & (n_1+n_2+n_5)\mathbf{I}_3 & -n_2\mathbf{I}_3 & -n_5\mathbf{I}_3 \\ -n_6\mathbf{I}_3 & -n_2\mathbf{I}_3 & (n_2+n_3+n_6)\mathbf{I}_3 & -n_3\mathbf{I}_3 \\ -n_4\mathbf{I}_3 & -n_5\mathbf{I}_3 & -n_3\mathbf{I}_3 & (n_3+n_4+n_5)\mathbf{I}_3 \end{bmatrix}. \quad (18)$$

We assume that the initial stresses on each triangular part of the quadrangular element are defined according to the same strategy described above, for the triangular NFDM element, considering the same director plane, intersecting each of the four triangular parts. Fig. 9 displays the local coordinates system considered for each part of the quadrangular element. Once the initial stresses on each triangular part are defined, the natural forces acting along the sides of each part are given by

$$\mathbf{N}_\alpha = V_\alpha \mathcal{L}_\alpha^{-1} \mathbf{T}_\alpha^{-T} \hat{\boldsymbol{\sigma}}_\alpha, \quad (15)$$

where index α spans the four element parts (*a,b,c,d*).

The natural force densities in each part $\mathbf{n}_\alpha = [n_{1\alpha} \quad n_{2\alpha} \quad n_{3\alpha}]^T$, depicted in Fig. 10, are now collected into four force density vectors, according to

$$\mathbf{n}_\alpha = \mathcal{L}_\alpha^{-1} \mathbf{N}_\alpha = V_\alpha \mathcal{L}_\alpha^{-2} \mathbf{T}_\alpha^{-T} \hat{\boldsymbol{\sigma}}_\alpha, \quad (16)$$

where $\mathcal{L}_\alpha = diag\{\ell_{1\alpha},\ell_{2\alpha},\ell_{3\alpha}\}$ and $\hat{\boldsymbol{\sigma}}_\alpha$ is the 2PK stress vector at the reference configuration of each part, expressed as a pseudo-vector.

Finally, we define a new vector of the natural force densities for the quadrangular element as

$$n = \begin{bmatrix} n_1 & n_2 & n_3 & n_4 & n_5 & n_6 \end{bmatrix}^T, \quad (17)$$

where $n_1 = n_{2a} + n_{1c}$, $n_2 = n_{1b} + n_{2c}$, $n_3 = n_{2b} + n_{1d}$, $n_4 = n_{1a} + n_{2d}$, $n_5 = n_{3a} + n_{3b}$, $n_6 = n_{3c} + n_{3d}$.

Once again we arrive at a linear relationship between internal forces and node coordinates, according to(3), where now the element nodal coordinates are given by $\mathbf{x} = [\mathbf{x}_1^T \quad \mathbf{x}_2^T \quad \mathbf{x}_3^T \quad \mathbf{x}_4^T]^T$, and where the *stiffness matrix* of the NFDM for the quadrangular element is given by

When all the element contributions are assembled, we again arrive at a linear problem at a global level, and once the viable geometry is found, the associated Cauchy stresses on each element part are obtained by inverting Eq. (16), according to:

$$\hat{\boldsymbol{\sigma}}_\alpha = \mathbf{T}_\alpha^T \boldsymbol{L}_\alpha^2 V_\alpha^{-1} \mathbf{n}_\alpha, \quad (19)$$

Finally, the stresses on the quadrangular element are obtained by an average of the stresses in each part, transformed to the coordinate system of the quadrangular element, and weighted by the corresponding areas, according to

$$\bar{\boldsymbol{\sigma}} = \frac{1}{A} \sum_{\alpha=1}^{4} A_\alpha \mathbf{R}_\alpha^T \boldsymbol{\sigma}_\alpha \mathbf{R}_\alpha \quad (20)$$

where $A = \sum_{\alpha=1}^{4} A_\alpha$ and $\mathbf{R}_\alpha$ are rotation matrices from the local coordinate basis of each part to the element coordinate basis (as defined in Fig. 9), given by

$$\mathbf{R}_\alpha = [\mathbf{e}_{\tilde{x}\alpha} \quad \mathbf{e}_{\tilde{y}\alpha} \quad \mathbf{e}_{\tilde{z}\alpha}]^T [\mathbf{e}_{\hat{x}} \quad \mathbf{e}_{\hat{y}} \quad \mathbf{e}_{\hat{z}}] \quad (21)$$

Our numerical tests indicate that the effect of the nonconformity of the two layers of the element on the final shape is negligible and, although somewhat arbitrary, the average stresses provided by Eq. (20) is also quite satisfactory, as suggested by the applications presented in the sequel.

5. Applications

The performance of the original triangular NFDM element was assessed on previous publications (Pauletti, 2006, 2008, 2011). In

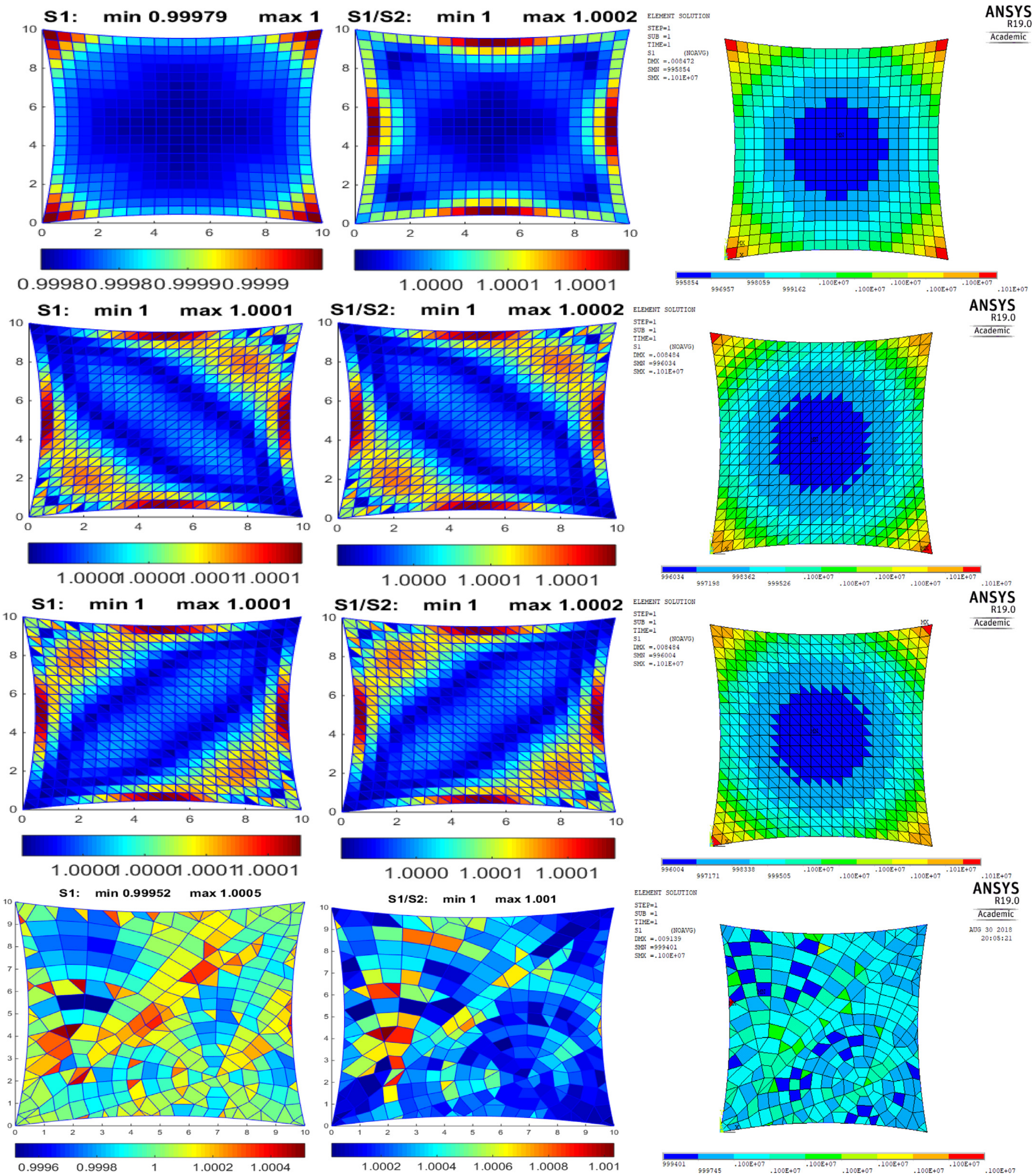

Fig. 16. Stress field results for Models 1 to 4 (from line 1 to 4) after 10 NFDM step . 1st column: 1st principal stresses σ_I; 2nd column: stress ratio σ_I/σ_{II}; 3rd column 1st principal stress σ_I obtained with Ansys.

this paper, we investigate the performance of the new quadrangular element, showing that it overcomes the structural bias introduced by triangular tessellations. We also explore the use of irregular tessellations, freely comprising quadrangular and triangular elements.

We initially consider a flat square, reference domain divided into different tessellations. Fig. 11(a) shows the refence domain regularly divided into a set of quadrangular elements ('Model 1'). Fig. 11(b) shows the same domain regularly divided into a set of triangular elements ('Model 2'). The quadrangular mesh of Model

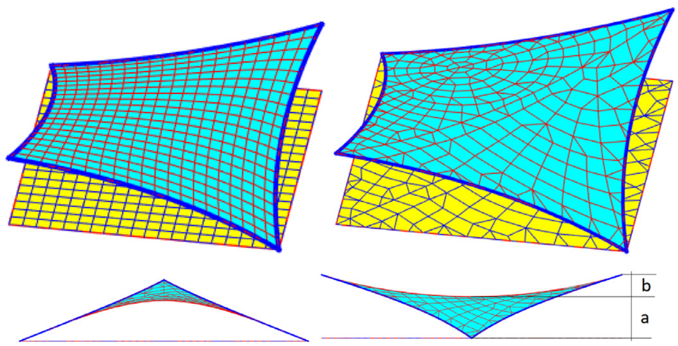

Fig. 17. Viable configuration for orthotropic stress fiel with $a/b > 1$.

1 preserves symmetry, if the mesh is reflected or rotated by 90°, but a bias is introduced by the orientation of the triangular mesh of Model 2. Therefore, we consider also the mesh obtained by the reflection of Model 2 as a different numerical model ('Model 3'). We further consider the reference domain divided into an irregular tessellation, shown in Fig. 11(c) ('Model 4'), which purposely preserves the topology of the FDM network previously considered in Fig. 3(a). We remark that the mesh of Model 4 is considerably coarser than the other models, thus coarser numerical results are to be expected. We consider that in all the cases the domain is bounded by standard FDM elements.

An interesting property of the NFDM is that the same refence mesh can be deformed into a broad range of shapes, with considerable freedom. shows Models 1 and 4 deformed in a single NFDM step into different shapes, simply by prescribing different set of nodal displacements, alongside with a uniform and isotropic 2PK stress field $\hat{\boldsymbol{\sigma}}_0 = [1 \quad 1 \quad 0]^T$ and a uniform initial normal load N_0 at the border cables. The saddle geometry shown at the first column of Fig. 12 was obtained with $N_0 = 1$, whereas the geometry shown at the column 2 was obtained with the same displacements, but with $N_0 = 30$. Discrepancies between results of the four models are negligible, both in terms of displacements and stresses, thus we show only results for Models 1 and 4. Some stress concentrations are verified in the regions where the elements undergo larger deformation, a property intrinsic to both FDM and NFDM methods, but overall, the final geometry and the associated stress field of NFDM are not substantially affected by the lay-out of the meshes. We provide quantitative comparisons on the resulting stress fields made in the benchmarks present in the sequel.

5.1. A saddle membrane generated in a single NFDM step

Fig. 13 shows the geometry of a saddle membrane surface generated in a single step of the NFDM, applying an initial uniform isotropic stress field $\hat{\boldsymbol{\sigma}}_0 = [1 \quad 1 \quad 0]^T$ to a $10m \times 10m$ square reference domain, alongside with a uniform normal load $N_0 = 30$ on the border cables, and equal vertical displacements $u_z = 3m$ to two opposite corners, while the other two corners are fixed. Models 1, 2 and 3 share the same set of nodes, and all produce the same displacements, thus only the geometry provided by Model 1 is compared to the geometry provided by the irregular tessellation (Model 4). Although these models have two different set of nodes, their final surface geometries are overall coincident.

The first column of Fig. 14 compares the fields of 1st principal Cauchy stresses σ_I for the four models. The resulting stress fields are very similar, with only slight differences on the maximum stress values. The quadrangular tessellation (Model 1) presents equal maximum 1st principal stress $\sigma_I = 1.39$ at the four vertices, whereas the two triangular tessellations (Models 2 and 3) present different maximum 1st principal stresses $\sigma_I = 1.39$ and $\sigma_I = 1.43$ at alternate vertices. These maxima occur at the vertices to which only one triangular element is connected. Results obtained with the irregular tessellation (Model 4) are shown at the fourth line. It is seen that the results are in fair agreement with the other models, with a maximum 1st principal stress $\sigma_I = 1.434$ at the corners. The second column of Fig. 14 displays the fields of 2nd principal Cauchy stresses σ_{II}, and again results are in close agreement between the four models. We remark that since the resulting stress fields are non-uniform, the configurations provided by this one step of the NFDM, although viable, are not minimal ones.

To confirm the quality of the configurations provided by a single step of the NFDM, the results obtained with the four models (*i.e.* final geometry, stresses on membrane and normal loads on border cables) were input as initial configurations to the Ansys FEM program, and non-linear equilibrium analyses were performed, assuming a flexible membrane (SHELL181) bounded by flexible cables (LINK180). The rationale for this test is that no significant displacements or stress variations should be verified on Ansys, if the NFDM final configurations are in fact good ones. We remind that force densities methods are based solely in equilibrium, whilst Ansys requires definition of material properties. Therefore, we considered realistic stiffness moduli $E \times t = 1.0$MN/m for the membrane and $E \times A = 6.84$MN for the border cables, and we interpreted stresses given by the NFDM as given in MPa.

The maximum displacements obtained with Ansys (0.0105 m in the cases of Models 1, 2 and 3, and 0.0096 m int the case of Model 4) are very small, if compared to the dimensions of the models and are basically associated to small deformations of the border cables. These residual deformations could be further reduced, by increasing the axial stiffness of the cables. The third column of Fig. 14 displays the σ_I fields obtained with Ansys. Excellent agreement is found between stresses previously calculated with NFDM, corroborating their quality. The maximum σ_I stresses calculated by Ansys occurs at the same vertices as before, with values of {1.39; 1.44; 1.44; 1.43} for Models 1 to 4, respectively. Visual discrepancies between NFDM and Ansys stress field renderings are basically due to the differences between our in-house plotting routines and those intrinsic to Ansys.

5.2. A minimal membrane surface generated with multiple NFDM steps

We further consider the use of the four models of Fig. 11 for the shape finding of a minimal surface, to which the NFDM converges after a few steps. Fig. 15 shows the results of re-imposing a uniform and isotropic 2PK stress field $\hat{\boldsymbol{\sigma}}_0 = [1 \quad 1 \quad 0]^T$ on the membrane, as well as a uniform normal load $N_0 = 30$ on the border cables, considering the shapes obtained in 10 subsequent steps of the NFDM. Once again, we impose equal vertical displacements $u_z = 3m$ to two opposite corners, while the other two corners are fixed. The four models converge to the same surface geometry; thus, we compare only the geometries provided by Models 1 and 4.

The first column of Fig. 16 shows the fields of 1st principal Cauchy stress σ_I for the four models. The stress ranges are very narrow, $0.99979 \leq \sigma_I \leq 1.00000$ on Model 1 and $1.00000 \leq \sigma_I \leq 1.00010$ on Models 2 and 3. Model 4 has a coarser mesh, but still a very narrow stress range, $0.99952 \leq \sigma_I \leq 1.00050$. The patchwork appearance of the plotting reflects the irregularity of the mesh. The second column of Fig. 16 displays the fields of the stress ratio σ_I/σ_{II}, which also presents a very narrow range around unity ($1 \leq \sigma_I/\sigma_{II} \leq 1.0002$ for Models 1, 2 and 3 and $1 \leq \sigma_I/\sigma_{II} \leq 1.0010$ for Model 4). Therefore, all four models converged to the minimal geometry, confirming our intuitive assumptions on the performance of the quadrangular element, with respect to the two alternative triangular partitions. Moreover, results are once again vali-

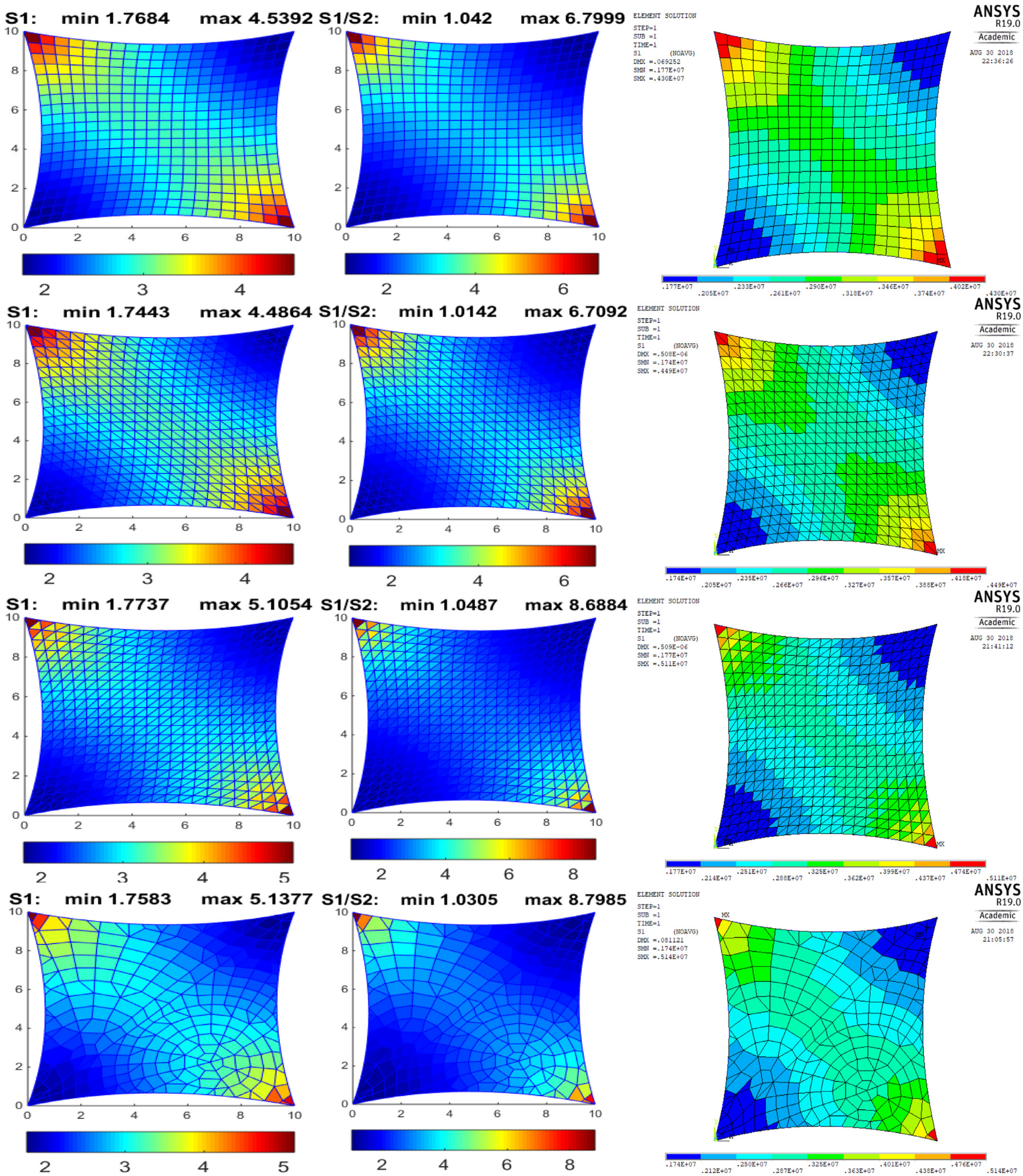

Fig. 18. Stress fields onto a non-minimal surface, for Models 1 to 4 (from line 1 to 4), after 10 NFDM step 1st column: 1st principal stresses σ_I; 2nd column: stress ratio σ_I/σ_{II}; 3rd column 1st principal stress σ_I, obtained with Ansys.

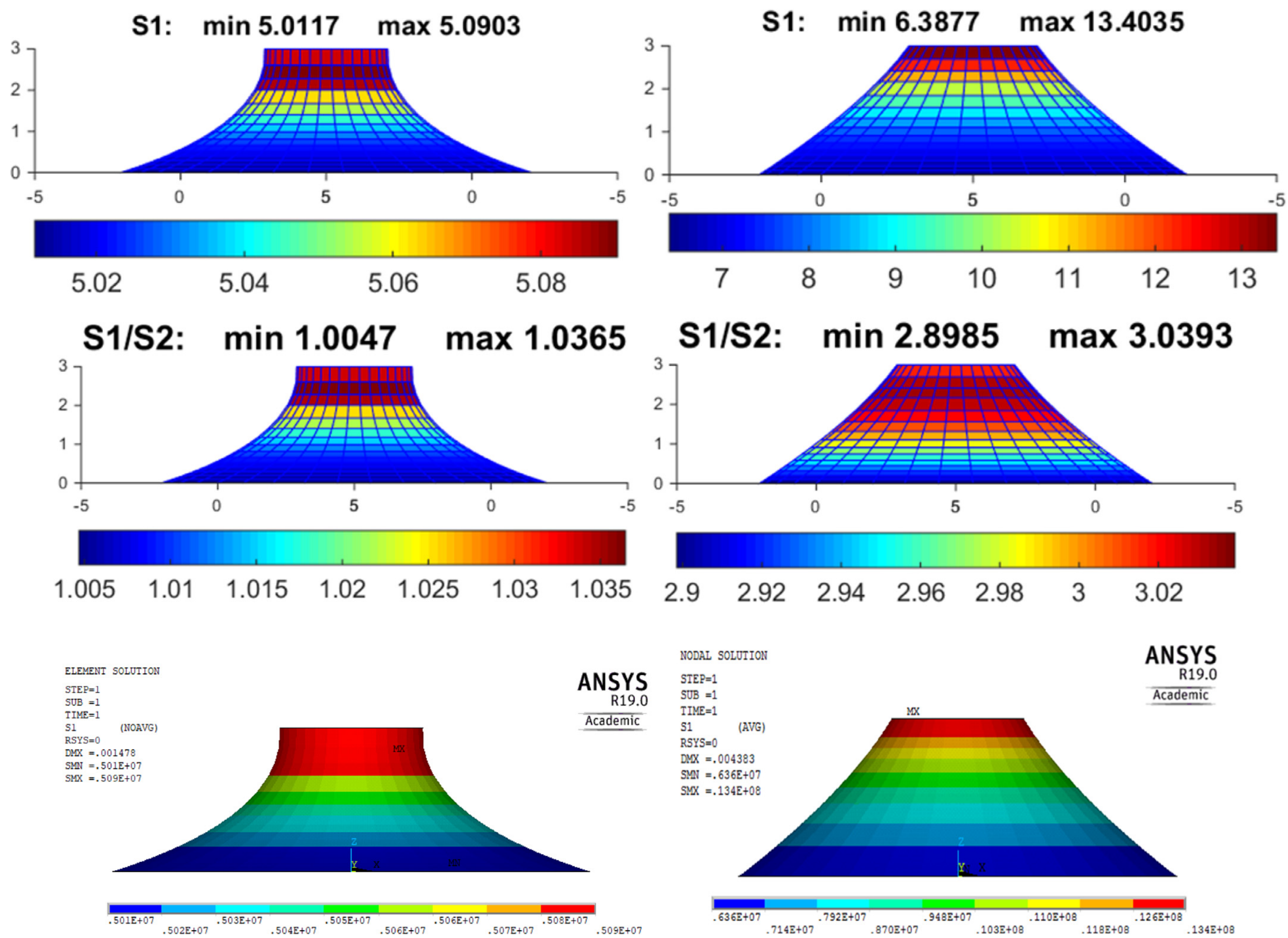

Fig. 19. Non-minimal (first column) and minimal (second column) conoid, obtained using NFDM and Ansys The first line shows the 1st principal Cauchy stress (σ_I) fields obtained with NFDM. The second line shows the ratios σ_I/σ_{II}, again with NFDM. The third line shows $\sigma_I$fields obtained with Ansys.

dated by Ansys software, which presents negligible displacements, when the NFDM final configurations are used as input for nonlinear equilibrium analyses (maximum 0.00848 m, for Models 1 to 3 and 0.009139 m for Model 4). The stress ranges obtained with Ansys are also in fair agreement ($0.996 \le \sigma_I \le 1.01$ for Models 1, 2 and 3, and $0.999 \le \sigma_I \le 1.00$ for Model 4). It can also be seen that the alternative triangular tessellations introduce only slight biases on the stress results, which, negligible as they may be regarded, are fully removed with the use of quadrangular elements. These biases are almost imperceptible in the Ansys results, which also smooth out the slight stress concentrations along the borders, accommodated by some small deformation along the border cables. Besides, the patchwork appearance of the σ_I stress field is smoothed, due to small deformations of the membrane elements.

5.3. Controlling geometry through prescription of an orthotropic stress field

We further employ the four numerical models presented above to the shape-finding of non-minimal surfaces under an initial uniform orthotropic 2PK stress fields with $\sigma_I = 3$ and $\sigma_{II} = 1$ which allows control of the height of the saddle point of the resulting surface, an important parameter for practical design applications, since it allows to adjust the structure's ceiling clearance, as suggested in Fig. 17. We have considered director planes parallel to the ridge diagonal, and we have imposed σ_I stresses parallel to the intersection of the director planes with each element surface, at each iteration.

After one iteration we obtain $a = 1.97$ and $b = 1.03$ for all the four models. Different ratios σ_I/σ_{II} could be tried to control the ratio a/b . The first column of Fig. 18 shows that Model 1 provides a 1st principal stress range of $1.768 \le \sigma_I \le 4.539$, with intermediate values regarding Models 2 and 3, which yield $1.744 \le \sigma_I \le 4.486$ and $1.774 \le \sigma_I \le 5.105$, respectively. Model 4 displays a stress range $1.758 \le \sigma_I \le 5.138$, a result closer to Model 3, because of the similar arrangement of the elements at the more stressed corners.

The second column shows that in Model 1 the stress ratio varies in the range $3 < \sigma_I/\sigma_{II} \le 6.80$, along the ridge diagonal (connecting the two high vertices), and in the range $1.04 \le \sigma_I/\sigma_{II} < 3$ along the valley diagonal (connecting the two low vertices). In Models 2 and 3, we have respectively ranges of $3 < \sigma_I/\sigma_{II} \le 6.71$ and $3 < \sigma_I/\sigma_{II} \le 8.69$, along the ridge diagonal, and ranges of $1.014 \le \sigma_I/\sigma_{II} < 3$ and $1.049 \le \sigma_I/\sigma_{II} < 3$, along the valley diagonal. Besides, Model 4 displays stress ratio ranges of $3 < \sigma_I/\sigma_{II} \le 8.80$ along the ridge diagonal and $1.03 \le \sigma_I/\sigma_{II} < 3$ along the valley diagonal. It is seen thus that, in this example, a somewhat higher bias is induced by the orientation of the triangular tessellation.

The third column of Fig. 18 presents the 1st principal stress fields obtained using Ansys ($1.77 \le \sigma_I \le 4.30$for Model 1, $1.74 \le \sigma_I \le 4.49$ for Model 2, $1.77 \le \sigma_I \le 5.11$for Model 3, and $1.74 \le$

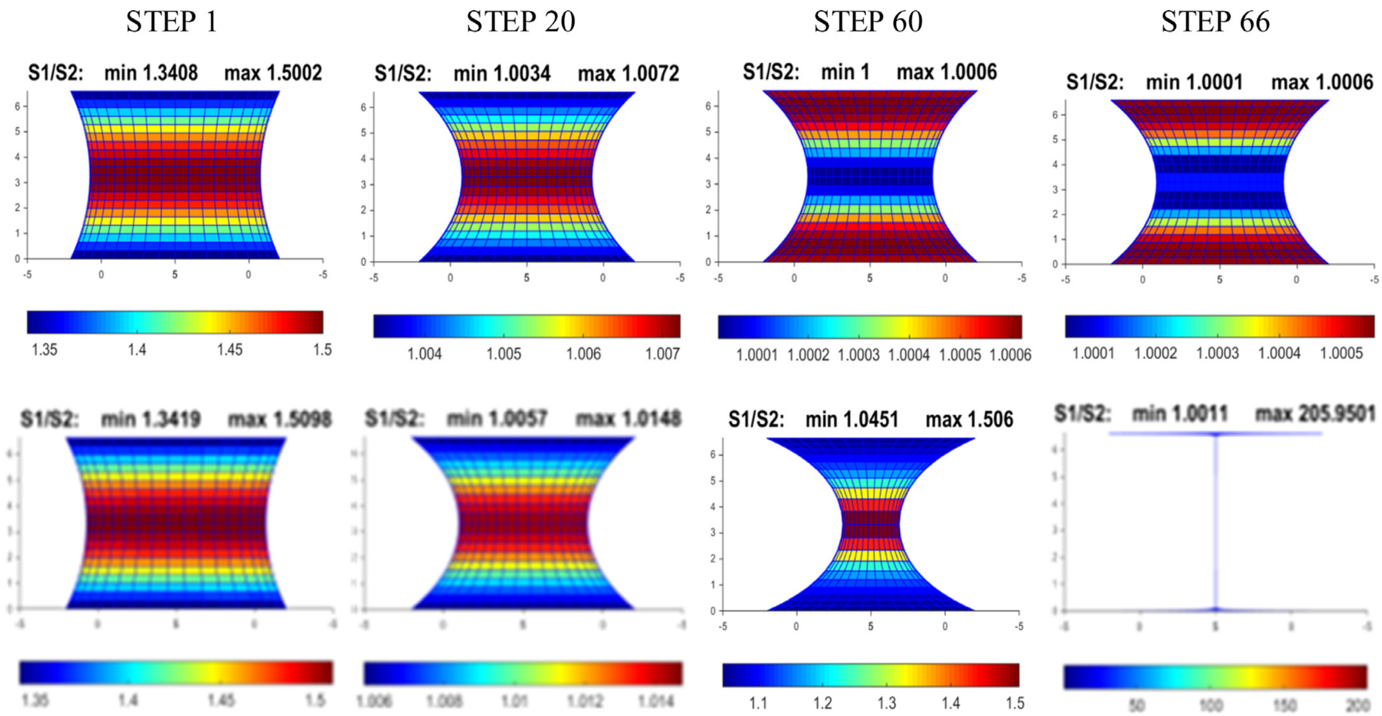

Fig. 20. Several steps of NFDM to find the minimal catenoid, with $h = 1.32R$(fist line) and $h = 1.33R$ second line). The stress ratioσ_I/σ_{II} is plotted over the resulting geometry at each step. .

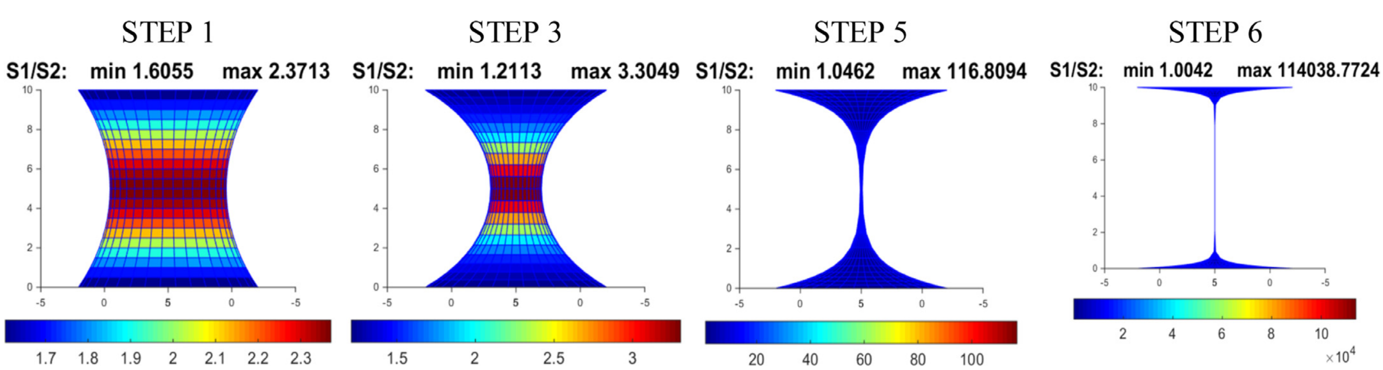

Fig. 21. Several steps of NFDM to find the minimal catenoid, with $h = 2.0R$. The stress ratio σ_I/σ_{II} is plotted onto the resulting geometry at each step. Necking is already clear a the 3rd step, and divergence occurs after very few iterations.

$\sigma_I \leq 5.14$ for Model 4), quite small variations with respect to the corresponding NFDM results of column 1. Maximum displacements are also very small, if compared to the dimensions of the models (0.07 m for Model 1, 5×10^{-7}m for Models 2 and 3, and 0.08 m for Model 4), once again validating the NFDM results. Once again, we remark that visual discrepancies between NFDM and Ansys renderings are basically due to the differences between our in-house plotting routines and those intrinsic to Ansys.

5.4. Comparison between minimal and non-minimal conoids

The first column of Fig. 19 compares a minimal conoid (stress ratio$\sigma_I/\sigma_{II} = 1$), to a non-minimal conoid (stress ratio$\sigma_I/\sigma_{II} = 3$), shown it the 2nd column, after 5 steps of the NFDM, using a regular quadrangular mesh. Geometry and stress results agree well with those given by Pauletti (2011) and Gellin and Pauletti (2013), where triangular meshes were considered. The first line of Fig. 19 shows the 1st principal stress fields $\sigma_I$on both conoids. The second line confirms that the condition $\sigma_I/\sigma_{II} = 3$ is fairly achieved after 5 iterations. The third line shows the 1st principal stress on both models as computed independently by Ansys, displaying satisfactory match with NFDM results.

5.5. Checking Goldschmidt limit for straight catenoids

As a final example, we consider the shape finding of a catenoid with equal lower and upper radiuses, separated by a given height. The benchmark is interesting, since minimal shapes are only possible for ring separations bellow the Goldschmidt limit (Goldschmidt, 1831), $h_{\text{lim}} = 1.3254868\,R$. Above this limit, the minimal solution corresponds to two disjoint flat surfaces, bounded by the two rings, and recursive imposition of isotropic stresses must lead to progressive mesh distortion, up to numerical divergence.

The first line of Fig. 20 shows several steps of the NFDM, for rings separated by a height $h = 1.32R$, just below Goldschmidt limit. Recursive imposition of a uniform and isotropic 2PK stress field converges to the expected catenoid geometry in a few steps. On the contrary, for $h = 1.33R$, just above Goldschmidt limit, successive iterations display incremental distortion, up to degeneration of the mesh, as shown in the second line of Fig. 20. For larger ring separations, divergence occurs in fewer steps. Fig. 21 shows results for $h = 2.0R$. A pronounced necking is perceived at step 3, steeply rising until divergence, after step 6. We remark that the verification of Goldschmidt limit is a clear indication of the physical soundness of the NFDM. Nonetheless, the shapes generated by the method, even above this limit are in fact viable shapes, although non-minimal, and could perhaps be selected for practical design purposes.

6. Conclusions

This paper outlined the formulation of the Natural Force Density Method for the shape finding of funicular structures (*i.e.*, cable nets, membranes and bending-free shells), proposing a more concise deduction of its main equations, as compared to previous works. The paper shows that NFDM offers clear advantages with respect to the original Force Density Method, to cope with irregular meshes.

Although FDM networks are routinely used in design practice to the find the shape of funicular surfaces, it may become quite dubious which force densities should be prescribed to achieve a desired configuration, comprising a geometric shape and an associate stress field. This is seldom remarked in papers about the method, and we expect that the current publication contributes to increase awareness on this issue.

The paper also presents an extension of NFDM to quadrangular elements, an important feature that allows a broader range of meshes to be used without prior adaptation. The paper shows that the proposed quadrangular element overcomes the bias that triangular meshes might introduce on the stress results. Some selected benchmarks are presented, indicating that the quadrangular element has excellent performance for shape finding of funicular structures, of either minimal or non-minimal surfaces, both in terms of the quality of the shapes as in terms of quality of the associated stress fields.

Appendix – Deduction of Eq. (8)

The internal nodal loads acting on a triangular membrane element under a uniform plane stress field $\boldsymbol{\sigma}$ can be assembled according either to Eqs. (6) or (7). Comparing these equations, we write down

$$\mathbf{p} = \begin{bmatrix} N_2\mathbf{v}_2 - N_3\mathbf{v}_3 \\ N_3\mathbf{v}_3 - N_1\mathbf{v}_1 \\ N_1\mathbf{v}_1 - N_2\mathbf{v}_2 \end{bmatrix} = \frac{t}{2}\begin{bmatrix} \ell_2\boldsymbol{\rho}_2 + \ell_3\boldsymbol{\rho}_3 \\ \ell_1\boldsymbol{\rho}_1 + \ell_3\boldsymbol{\rho}_3 \\ \ell_1\boldsymbol{\rho}_1 + \ell_2\boldsymbol{\rho}_2 \end{bmatrix} \tag{A1}$$

where $\boldsymbol{\rho}_i = \boldsymbol{\sigma}\mathbf{n}_i$, are the stress vectors acting along the sides of the element, $\boldsymbol{\sigma}$ is the stress tensor, $\mathbf{n}_i$ are unit vectors normal to the element sides and $\mathbf{v}_i$ are unit vectors parallel to the element sides, as depicted in Fig. 6. Expressing these quantities in the element local coordinate system, the components of all involved quantities in the $\hat{z}$ direction (normal to the element surface) are null, so we reduce the problem to two dimensions and, marking the quantities with un upper hat, to represent the adopted coordinate system, we write explicitly:

$$\hat{\boldsymbol{\sigma}} = \begin{bmatrix} \sigma_x & \tau_{xy} \\ \tau_{xy} & \sigma_y \end{bmatrix}; \hat{\boldsymbol{n}}_1 = \begin{bmatrix} \sin\gamma \\ \cos\gamma \end{bmatrix}; \hat{\boldsymbol{n}}_2 = \begin{bmatrix} -\sin\beta \\ \cos\beta \end{bmatrix};$$

$$\hat{\boldsymbol{n}}_3 = \begin{bmatrix} 0 \\ -1 \end{bmatrix}; \hat{\boldsymbol{v}}_1 = \begin{bmatrix} -\cos\gamma \\ \sin\gamma \end{bmatrix}; \hat{\boldsymbol{v}}_2 = \begin{bmatrix} -\cos\beta \\ -\sin\beta \end{bmatrix}; \hat{\boldsymbol{v}}_3 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \tag{A2}$$

Considering the internal loads acting on node 1, we have a set of two equations on the unknowns N_2 and N_3:

$$N_2\cos\beta + N_3 = \frac{t}{2}((\ell_2\sin\beta)\,\sigma_x + (\ell_3 - \ell_2\cos\beta)\,\tau_{xy}) \tag{A3}$$

$$N_2\sin\beta = \frac{t}{2}((\ell_3 - \ell_2\cos\beta)\,\sigma_y + \ell_2\sin\beta\,\tau_{xy}) \tag{A4}$$

Focusing on Eq. (A4), we use the law of sines $\ell_2 \sin\alpha = \sin\gamma\,\ell_3$ and consider that $\sin\alpha = \sin\gamma\cos\beta + \cos\gamma\sin\beta$ (since $\alpha + \beta + \gamma = \pi$). Then, after some manipulation, we obtain

$$N_2 = \frac{t\ell_3}{2\sin\alpha}(\cos\gamma\,\sigma_y + \sin\gamma\,\tau_{xy}) \tag{A5}$$

Now, the volume of the triangular element depicted in Fig. 5 can be calculated by

$$V = \frac{t\ell_2\ell_3\sin\beta}{2} \tag{A6}$$

Substituting (A6) into (A5) we arrive at

$$N_2 = \frac{V}{\ell_2}\left(\frac{\cos\gamma}{\sin\alpha\sin\beta}\sigma_y + \frac{\sin\gamma}{\sin\alpha\sin\beta}\tau_{xy}\right) \tag{A7}$$

With similar manipulations, we obtain

$$N_3 = \frac{V}{\ell_3}\left(\sigma_x - \frac{\cos\beta\cos\gamma}{\sin\beta\sin\gamma}\sigma_y + \frac{\cos\gamma\sin\beta - \sin\gamma\cos\beta}{\sin\beta\sin\gamma}\tau_{xy}\right) \tag{A8}$$

Furthermore, considering any one of the four equations provided by the equilibrium of nodes 2 and 3, we obtain, after some further manipulation

$$N_1 = \frac{V}{\ell_1}\left(\frac{\cos\beta}{\sin\gamma\sin\alpha}\sigma_y - \frac{\sin\beta}{\sin\gamma\sin\alpha}\tau_{xy}\right) \tag{A9}$$

The redundant equations reflect the reciprocity of the forces acting on nodes of same side. Collecting (A9), (A5) and (A8) into a single matrix equation we have:

$$\begin{bmatrix} N_1 \\ N_2 \\ N_3 \end{bmatrix} = \begin{bmatrix} \frac{V}{\ell_1} & 0 & 0 \\ 0 & \frac{V}{\ell_2} & 0 \\ 0 & 0 & \frac{V}{\ell_3} \end{bmatrix} \begin{bmatrix} 0 & \frac{\cos\beta}{\sin\gamma\sin\alpha} & -\frac{\sin\beta}{\sin\gamma\sin\alpha} \\ 0 & \frac{\cos\gamma}{\sin\beta\sin\alpha} & \frac{\sin\gamma}{\sin\beta\sin\alpha} \\ 1 & -\frac{\cos\beta\cos\gamma}{\sin\beta\sin\gamma} & \frac{\sin\beta\cos\gamma - \cos\beta\sin\gamma}{\sin\beta\sin\gamma} \end{bmatrix} \times \begin{bmatrix} \sigma_x \\ \sigma_y \\ \tau_{xy} \end{bmatrix} \tag{A10}$$

Or, more compactly

$$\mathbf{N} = V\boldsymbol{\mathcal{L}}^{-1}\mathbf{T}^{-T}\hat{\boldsymbol{\sigma}}, \tag{A11}$$

where we have implicitly defined the natural loads vector $\mathbf{N}$, the length matrix $\boldsymbol{\mathcal{L}}$, the transformation matrix $\mathbf{T}^{-T}$and the pseudo-vector $\hat{\boldsymbol{\sigma}}$, which collects the independent components of the stress tensor. Eq. (A11) corresponds to Eq. (8), in the body of this paper.

We remark that the somewhat award notation for the transformation matrix $\mathbf{T}^{-T}$ defined by (A10) and (A11) is a reminiscence of the original deduction of the NFDM (Pauletti, 2006), were we first defined its transpose-inverse

$$\mathbf{T} = \begin{bmatrix} \cos^2\gamma & \cos^2\beta & 1 \\ \sin^2\gamma & \sin^2\beta & 0 \\ -\sin\gamma\cos\gamma & \sin\beta\cos\beta & 0 \end{bmatrix}, \tag{A12}$$

which related some *natural strains* (*i.e.* parallel to the sides of the element), to Green's strains according to $\boldsymbol{\varepsilon}_n = \mathbf{T}\hat{\boldsymbol{\varepsilon}}$. Then we sought some energetically-conjugate *natural stresses*$\boldsymbol{\sigma}_n$, by the principle of virtual work, such that $\delta\hat{\boldsymbol{\varepsilon}}^T\hat{\boldsymbol{\sigma}} = \delta\boldsymbol{\varepsilon}_n{}^T\boldsymbol{\sigma}_n$, $\forall\delta\hat{\boldsymbol{\varepsilon}}$. Integration of these natural stresses again provided the internal force vector $\mathbf{p}$, and finally, once again Eq. (A11). However, the NFDM is unmaterial, *i.e.*, based solely on equilibrium conditions, and the definition of natural strains and stresses is altogether unnecessary for setting up the method.

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